



Summary Statistics in Public Health

This course is part of [Biostatistics in Public Health Specialization](#)

 Instructor: [John McGready, PhD, MS](#)

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6 modules
Gain insight into a topic and learn the fundamentals.

4.8 ★
2062 reviews

Beginner level
Recommended experience ⓘ

Flexible schedule
2 weeks at 10 hours a week
Learn at your own pace

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What you'll learn

- ✓

Calculate continuous data measures
- ✓

Interpret data visualizations
- ✓

Analyze binary data
- ✓

Analyze time to event data

Skills you'll gain

- Sampling (Statistics)
- Risk Analysis
- Statistical Visualization
- Data Literacy
- Statistical Methods
- Statistics
- Quantitative Research
- Probability Distribution
- Analysis
- Statistical Inference
- Show all

Details to know

- 

Shareable certificate
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Assessments
10 assignments
- 

Taught in English
[22 languages available](#)
- 

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Build your subject-matter expertise

This course is part of the [Biostatistics in Public Health Specialization](#)
When you enroll in this course, you'll also be enrolled in this Specialization.

- Learn new concepts from industry experts
- Gain a foundational understanding of a subject or tool
- Develop job-relevant skills with hands-on projects
- Earn a shareable career certificate



There are 6 modules in this course

Biostatistics is the application of statistical reasoning to the life sciences, and it is the key to unlocking the data gathered by researchers and the evidence presented in the scientific literature. In this course, we'll focus on the use of statistical measurement methods within the world of public health research. Along the way, you'll be introduced to a variety of methods and measures, and you'll practice interpreting data and performing calculations on real data from published studies. Topics include summary measures, visual displays, continuous data, sample size, the normal distribution, binary data, the element of time, and the Kaplan-Meier curve.

[Show more](#)

The Role of Statistics in Public Health Research

Module 1 • 1 hour to complete

[Module details](#)

Within this learning module, you will learn how statistics are used within public heath research. Topics covered include study design types, data types, and data summarization. You will complete a practice quiz before completing a graded quiz.

What's included

 6 videos  1 assignment

Hide info about module content

 6 videos • Total 64 minutes

Welcome • 1 minute

Introduction and Overview • 11 minutes

Samples Versus Population • 11 minutes

Considerations with Regard to Study Design • 23 minutes

Data Types and Summarization • 9 minutes

Additional Examples • 10 minutes

 1 assignment

Practice Quiz: The Role of Statistics in Public Health Research • 0 minutes

Continuous Data Measures

Module 2 • 4 hours to complete

[Module details](#)

Module two involves several lectures, a practice quiz and a graded quiz. Topics include summary statistics, visual displays, role of sample size, and continuous data.

What's included

 6 videos  1 reading  2 assignments

Hide info about module content

 6 videos • Total 88 minutes

Introduction • 2 minutes

Useful Summary Statistics • 25 minutes

Visual Displays • 16 minutes

Instructor

Instructor ratings **4.8** ★ (739 ratings)



John McGready, PhD, MS
Johns Hopkins University
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The Role of Sample Size on Sample Based Estimates • 16 minutes

Comparing Distributions of Continuous Data • 16 minutes

Additional Examples • 12 minutes

 1 reading • Total 10 minutes

Summative Quiz Solutions • 10 minutes

 2 assignments • Total 140 minutes

Practice: Continuous Data Measures • 60 minutes

Continuous Data Measures • 80 minutes

The Normal Distribution [Module details ^](#)

Module 3 • 3 hours to complete

The focus of this module is on normal distribution. Topics covered include defining the standard normal distribution, and application of principles of normal distribution to sample data. There is a practice quiz where you can test your knowledge before completing the graded quiz.

What's included

 5 videos  2 assignments

Hide info about module content ^

 5 videos • Total 75 minutes

Introduction • 3 minutes

The Standard Normal Distribution Defined • 19 minutes

Applying the Principles of the Normal Distribution to Sample Data to Estimate Characteristics of Population Data • 26 minutes

What Happens When We Apply the Properties of the Normal Distribution to Data Not Approximately Normal: A Warning • 14 minutes

Additional Examples • 13 minutes

 2 assignments • Total 90 minutes

Practice: The Normal Distribution • 30 minutes

The Normal Distribution • 60 minutes

Binary Data [Module details ^](#)

Module 4 • 3 hours to complete

Module four covers binary data and its significance. In addition to lectures, you will complete a practice quiz and a graded quiz.

What's included

 6 videos  1 reading  2 assignments

Hide info about module content ^

 6 videos • Total 78 minutes

Introduction • 3 minutes

Binary Data: Definition and Summarization (Binomial Distribution, P-Hat, SD) • 11 minutes

Comparing Distributions of Binary Data: Risk Difference and Relative Risk • 23 minutes

Comparing Distributions of Binary Data: Odds Ratios • 16 minutes

A Brief Note About Ratios (Will Also Be Revisited Later in the Course) • 12 minutes

Additional Examples • 13 minutes

 1 reading • Total 10 minutes

Summative Quiz Solutions • 10 minutes

 2 assignments • Total 90 minutes

Practice: Binary Data • 30 minutes

Binary Data • 60 minutes

Dealing with the Element of Time [Module details ^](#)

Module 5 • 3 hours to complete

In module five, you will explore how time is defined and studied in relation to data and learn about the Kaplan-Meier curve. In addition to a practice quiz, you will complete a graded quiz and project.

What's included

 6 videos  1 reading  2 assignments

Hide info about module content ^

 6 videos • Total 85 minutes

Introduction • 3 minutes

Time to Event Data: Definition (Censoring) and Numerical Summary Measures (Incidence Rates) • 24 minutes

Numerically Comparing Groups on Time to Event Outcomes • 11 minutes

Summarizing Time-to-Event Data, Graphically • 28 minutes

Graphically Comparing Groups on Time to Event Outcomes • 9 minutes

Additional Examples • 10 minutes

 1 reading • Total 10 minutes

Summative Quiz Solutions • 10 minutes

 2 assignments • Total 60 minutes

Practice: Dealing with the Element of Time • 0 minutes

Dealing with Elements of Time • 60 minutes

Course Project [Module details ^](#)

Module 6 • 2 hours to complete

During this module, you get the chance to demonstrate what you've learned by putting yourself in the shoes of biostatistical consultant on two different studies, one about asthma medication and the other about self-administration of injectable contraception. The two research teams have asked you to help them interpret previously published results in order to inform the planning of their own studies.

What's included

 2 readings  1 assignment

Hide info about module content ^

 2 readings • Total 40 minutes

Biostatistical Consulting Project • 30 minutes

Course Project Quiz Solutions • 10 minutes

 1 assignment • Total 60 minutes



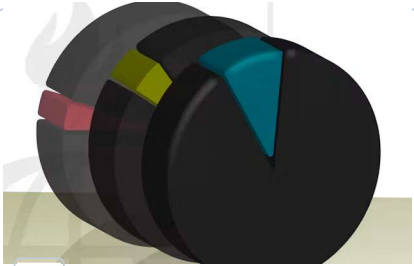
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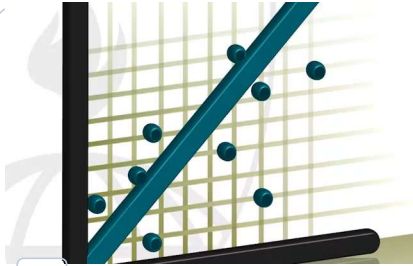
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Hypothesis Testing in Public Health

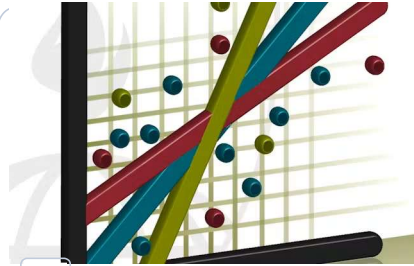
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Simple Regression Analysis in Public Health

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Multiple Regression Analysis in Public Health

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Learner reviews



Showing 3 of 2062

T

TN

★★★★★ 5.0 · Reviewed on Mar 26, 2020

Very good course! Lectures were easy to follow and gave great examples! Would highly encourage taking this course if you are looking for a biostatistics course or even to refresh your memory!

M

MP

★★★★★ 5.0 · Reviewed on Jul 5, 2022

it was an amazing opportunity for me to learn from a great man in the Biostatistics field and importantly from such a great university, all videos were so understandable and comprehensive.

M

MS

★★★★★ 5.0 · Reviewed on Feb 15, 2024

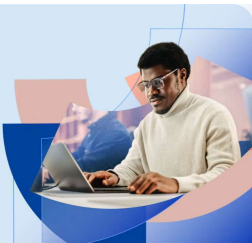
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